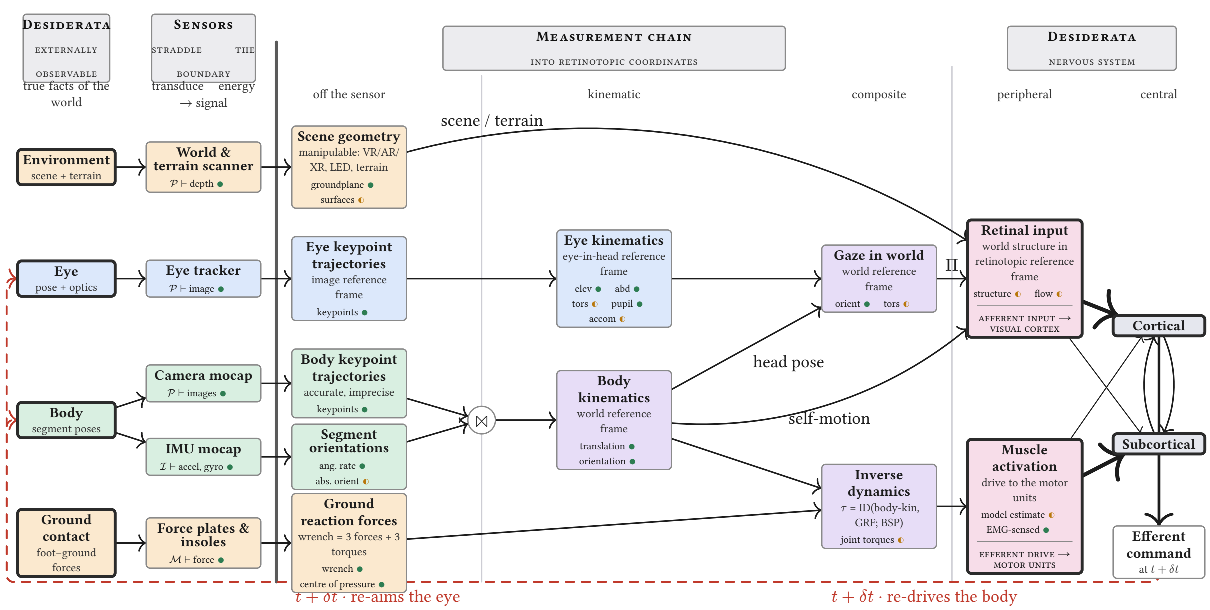
per channel — ● measured ◐ inferred under a prior · each estimate names the reference frame it lives in · ▮ the real–internal boundary (transduction)



previous instant ⟵ $t - \delta t$ · **this figure — a single instant t** · $t + \delta t$ ⟶ next instant *the efferent loop advances the frame by δt*

Reading the chart both ends are **desiderata** — the True Facts we want. The chain between them transports each estimate out of its sensor's reference frame and into the reference frame the nervous system uses.

Notation

- $\mathcal{E}$ an **energy** a sensor is sensitive to: $\mathcal{P}$ photons · $\mathcal{I}$ inertial · $\mathcal{M}$ contact
- $\mathcal{E} \vdash s$ **transduce** — a sensor **yields** ($\vdash$) a raw **signal** s; the one physical step, at the real–internal boundary
- $\rightarrow$ **derive** — a signal is processed into a measurement, then an estimate
- $\oplus$ **compose** — chain estimates across reference frames
- $\bowtie$ **fuse** — combine two estimates of one quantity
- Π **project** — map a state onto an observable

The reference-frame journey

Each estimate lives in one reference frame. The vision path is a journey from sensor reference frames into the retinotopic reference frame — the coordinates the visual cortex uses.

Vision image → eye-in-head → world → retinotopic
Body camera / IMU sensor → world (inertial)
Gaze eye-in-head ⊕ head-in-world → world

The chains — energy ⊢ signal → estimate

Eye	$\mathcal{P} \vdash$ image → keypoints → eye-kin (elev, abd, tors, pupil, accom)
Body	$\mathcal{P} \vdash$ images → keypoints; $\mathcal{I} \vdash$ accel+gyro → segment orient; body-kin = keypoints ⋈ orient
Retinal input	Π(gaze, scene, self-motion) — afferent
Motor	τ = ID(body-kin, GRF); muscle = M(τ) ⋈ EMG — efferent

Figure 1: **The measurement chain of a Dome.** The two ends are **desiderata** — the True Facts we want to know. On the left, the **externally observable** facts (the eye, the body, the environment); on the right, the facts that are **not** externally observable (the afferent retinal input, the efferent muscle activation, and the central nervous system). Between them sit the **sensors**, which straddle the real–internal boundary by transducing physical energy into signal (the heavy vertical rule), and the **measurement chain** that derives estimates from those signals. Its single job is to carry each estimate out of its sensor's reference frame and into the reference frame the nervous system uses: the vision path runs image ⟶ eye-in-head ⟶ world ⟶ **retinotopic**, the coordinates of the visual cortex. Every box is a bundle of **channels**, each marked measured (●) or inferred under a prior (◐), and every estimate names the reference frame it lives in. Fusion (⋈) and composition (⊕) build the kinematic and composite estimates; projection (Π) maps them onto the retinal input and the muscle activation. Those two peripheral signals drive the **central nervous system** (cortical and subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t**; the dashed red loop advances the frame by δt. The strip beneath the diagram gives the full reading rule, every symbol defined.